

ML Builder Expo

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone Train, Analyze, Improve

Without opening the old lesson, explain one key idea from Capstone Train, Analyze, Improve. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Model Cards and Human Review

Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace evidence for Machine learning project defense. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Machine learning project defense. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Review a defense claiming 95 percent success on 20 cases but omitting the baseline, split history and errors. Give three revision requests and explain why polish is insufficient.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Final counts are $TP=12, FN=3, FP=5, TN=20$. Compute accuracy, precision and recall, then state one limitation and a review trigger for the model defense.

6. Solve the given case

Final classifier gets 16/20 correct; two of four errors are false negatives.

Report supported metrics and what is missing.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Your expo model succeeds on three demonstration plants. What evidence should accompany the demo and what limitation should you state?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 36 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Deliver the final Responsible Plant Classifier product card, live evidence trace, model card, and human-oversight commitment.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Final counts are $TP=12, FN=3, FP=5, TN=20$. Compute accuracy, precision and recall, then state one limitation and a review trigger for the model defense.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Write an opening

Draft a three-sentence opening with purpose, a result and a limit.

My reasoning and evidence

13. W2 Trace evidence

Name the files needed to audit one reported accuracy figure.

My reasoning and evidence

14. W3 Answer critique

A peer asks whether your model works for every school. Give a supported response.

My reasoning and evidence

15. W4 Apply independently

Review a defense claiming 95 percent success on 20 cases but omitting the baseline, split history and errors. Give three revision requests and explain why polish is insufficient.

My reasoning and evidence

16. W5 Plan the next step

Write one next investigation tied to an observed limitation, and identify what must stay held out.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: ML Builder Expo

Present a plant-classifier expo claim that another team could check.

Toy expo evidence: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct. Three candidate failures are recorded; no outdoor trial exists.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

ML Builder Expo

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Show denominator

Candidate $12/15=80\%$; baseline $10/15\approx66.7\%$ on the same cases.



2. Show independence

Explain five held-out plant identities and three related views per plant.



3. Show failures

Present the three failure records and indoor-only limitation beside the headline rate.

CHANGE ONE THING

The poster omits the baseline and calls the model "field ready".

Explain the changed result using the matching lesson.

Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

